S.B. JAIN INSTITUTE OF TECHNOLOGY MANAGEMENT & RESEARCH, NAGPUR

Practical 11
Postlab (OPEN ENDED PRACTICAL)

Aim: To implement Round Robin CPU Scheduling algorithm and analyze its performance based on waiting time and turnaround time.

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- **Aim:** To implement Round Robin CPU Scheduling algorithm and analyze its performance based on waiting time and turnaround time.

- **Objectives:**
  - To understand Round Robin scheduling algorithm
  - To calculate Waiting Time and Turnaround Time
  - To analyze time-sharing scheduling technique

- **Requirements:**
  - ✓ **Hardware Requirements:**
    - Processor: Minimum 1 GHz
    - RAM: 512 MB or higher
    - Storage: 100 MB free space
  - ✓ **Software Requirements:**
    - Operating System: Linux/Unix-based
    - Compiler: GCC
    - Text Editor: Nano/Vim

- **Theory:**

Round Robin (RR) is a **preemptive CPU scheduling algorithm** designed for time-sharing systems. In this algorithm, each process is assigned a fixed time slice called **time quantum**.

Processes are executed in a circular queue. Each process gets CPU for a limited time. If it is not completed within that time, it is moved to the end of the queue.

This ensures fairness and avoids starvation.

Features of Round Robin:

- Fair allocation of CPU
- Preemptive in nature
- Suitable for time-sharing systems
- Reduces starvation

Limitations:

- Performance depends on time quantum
- Too small quantum → high context switching
- Too large quantum → behaves like FCFS

Important Terms

1. Waiting Time (WT)

Time a process spends waiting in ready queue

Formula:

WT = Turnaround Time – Burst Time

2. Turnaround Time (TAT)

Total time from arrival to completion

Formula:

TAT = Completion Time – Arrival Time

❖ CODE:

```
#include <stdio.h>

int main() {
    int n, tq;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    int bt[n], rt[n], wt[n], tat[n];

    printf("Enter Burst Time:\n");
    for(int i = 0; i < n; i++) {
        printf("P%d: ", i+1);
        scanf("%d", &bt[i]);
        rt[i] = bt[i]; // remaining time
    }

    printf("Enter Time Quantum: ");
    scanf("%d", &tq);

    int time = 0, remain = n;

    while(remain > 0) {
        for(int i = 0; i < n; i++) {
            if(rt[i] > 0) {
                if(rt[i] <= tq) {
                    time += rt[i];
                    wt[i] = time - bt[i];
                    rt[i] = 0;
                    remain--;
                } else {
                    time += tq;
                    rt[i] -= tq;
                }
            }
        }
    }

    int total_wt = 0, total_tat = 0;

    printf("\nProcess\tBT\tWT\tTAT\n");
    for(int i = 0; i < n; i++) {
        tat[i] = bt[i] + wt[i];
        total_wt += wt[i];
        total_tat += tat[i];

        printf("P%d\t%d\t%d\t%d\n", i+1, bt[i], wt[i], tat[i]);
    }

    printf("\nAverage WT = %.2f", (float)total_wt/n);
    printf("\nAverage TAT = %.2f", (float)total_tat/n);

    return 0;
}
```

❖ **Output:**

```
Enter number of processes: 5
Enter Burst Time:
P1: 4
P2: 3
P3: 10
P4: 5
P5: 2
Enter Time Quantum: 2

Process BT      WT      TAT
P1      4       8       12
P2      3       10      13
P3      10      14      24
P4      5       15      20
P5      2       8       10

Average WT = 11.00
Average TAT = 15.80

...Program finished with exit code 0
Press ENTER to exit console.
```

❖ **Conclusion**:
Round Robin scheduling is an efficient algorithm for time-sharing systems. It ensures fairness among processes by allocating equal CPU time. However, its performance depends heavily on the choice of time quantum. A properly selected time quantum improves system responsiveness and reduces waiting time.

❖ **Discussion Questions:**

1. **What is Round Robin scheduling?**
2. **What is time quantum?**
3. **Why is Round Robin called a preemptive algorithm?**
4. **What happens if time quantum is too large or too small?**
5. **How does Round Robin prevent starvation?What is the main difference between FCFS and SJF?**

❖ **References:**

https://www.geeksforgeeks.org/round-robin-scheduling/
https://chatgpt.com/

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Signature
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